ARTH.AI

The bank that understands why.

Causal, agentic AI for customer acquisition, digital adoption and engagement.

SBI BI Hackathon @ GFF 2026 | Theme: Agentic AI & Emerging Tech

Live prototype: arth-ai-one.vercel.app | Code: github.com/sahil0m/arth-ai

Right product. Wrong moment.

~60%	of people abandon a complex digital sign-up before they ever activate.
Too late	loan and product offers often arrive weeks after the need has passed.
Fixed calendar	banks engage on schedules, not when life actually creates a need.

The deeper issue: most banking AI predicts what a customer might do. It never understands why.

Predictive AI guesses. Causal AI understands.

CORRELATIONAL (what most banks do)

"Customers who buy jewellery often take loans."

Acts on the symptom. Fires for festival shoppers and the wealthy alike, and mistimes the offer.

CAUSAL (what ARTH.AI does)

"A wedding causes both the jewellery and the loan need."

Acts on the real cause and gets the timing right, with a plain reason for every decision.

This is the academic difference between prediction and causal / uplift modelling, applied to banking.

ARTH.AI

An agentic AI layer that reads a customer's transaction signals, works out the life event behind them, and acts at the right moment, in the right language, on the right channel.

Acquisition

Find and onboard the right customers

Digital adoption

Surface the right feature at the right time

Engagement

Act around real life events, not campaigns

One causal engine. All three pillars of the SBI mandate.

From a UPI transaction to a timed, explained action.

1

Data

consent-based UPI history

2

Signals

12 interpretable clues, scored 0-1

3

Causal engine

infer the hidden life event

4

Needs

rank the products it causes

5

Timing

the best day to reach out

6

Agents

take the action

Stage 3 is the core: a structural causal model inverted with Bayesian reasoning. Everything else feeds it or acts on it.

Four agents, one causal brain.

SPARSH
Acquisition

Finds and onboards new customers with a quick, consent-based video-KYC flow.

PRAGATI
Digital adoption

Unlocks features at a pace that matches each customer's comfort, nudging only at good moments.

BANDHAN
Engagement

Answers financial stress with support, not sales, and offers loyal customers better rates first.

GYAAN
Meta-learning

Learns across customers to give privacy-safe, peer-validated guidance.

It works, and we measured it.

~70%

causal need accuracy

~29%

pattern-based accuracy

+138%

relative uplift

65-87%

real life-event recall

All reproducible live in the browser. The causal model nearly triples the accuracy of the pattern-based approach.

> Validated formally in Python with DoWhy (Microsoft Research): in a controlled test, 97% of a naive correlation was confounding. Placebo and random-common-cause refutation tests both passed.

It reads a customer in 45 seconds.

Example: Aditya, 36, Delhi. Six months of UPI data goes in.

- Signals: jewellery, venue, salary change
- **Causal read: WEDDING, 80% confidence**
- Needs: joint account, personal loan
- Timing: act near day 39, not today
- Action: WhatsApp in Hindi, at 7 PM

Notice the restraint: a spurious signal fired, yet the wedding still won, and the system chose to wait for the right day rather than pounce.

Compliant by design, not by patch.

RBI FREE-AI

Built to all 7 sutras: trust, fairness, accountability, explainability and more.

DPDP Act 2023

Per-signal, revocable consent. Raw data never leaves the device.

Human in the loop

Every credit decision is approved by a person.

Explainable & fair

A plain reason for every action; causal logic avoids proxy bias.

This prototype uses only synthetic, consent-simulated data. No real customer data is involved.

A B2B platform that pays for itself.

How we earn

Annual platform licence or subscription.

A success-based share of the incremental products and loans it drives.

Paid integration, deployment and support.

Why it scales

One engine lifts acquisition, adoption and retention at once.

For a bank of SBI's scale, a small percentage gain is significant value.

Cloud-native, so it scales at low marginal cost, then extends to other banks and NBFCs.

Why this wins.

- First live causal inference in Indian customer banking, to our knowledge.
- Nearly 3x the accuracy of the usual pattern-based approach, measured and reproducible.
- All three pillars (acquisition, adoption, engagement) on one shared engine.
- Built around RBI FREE-AI and DPDP from the first line of code.
- A working, deployable prototype today, with an honest path to production.

From pilot to national scale.

3 months

Sandbox pilot

A/B test on a consented cohort inside SBI.

6 months

Regional scale

RCT-calibrated, measured uplift across regions.

12 months

National

Full rollout with on-device federated learning.

The first causally intelligent bank.

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Thank you.